

RELOCATING LOADER

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SYSTEMS PROGRAMMING

WHAT IS A RELOCATING LOADER?

- IT IS A SYSTEM PROGRAM THAT LOADS AN OBJECT PROGRAM INTO MEMORY AT A DIFFERENT ADDRESS THAN THE ONE ORIGINALLY SPECIFIED DURING ASSEMBLY.
- IT ADJUSTS ALL ADDRESS-DEPENDENT VALUES (SUCH AS INSTRUCTION OPERANDS AND DATA REFERENCES) SO THE PROGRAM CAN EXECUTE CORRECTLY FROM ITS NEW LOCATION.
- USES MODIFICATION (M) RECORDS IN THE OBJECT FILE TO DETERMINE WHICH ADDRESSES REQUIRE ADJUSTMENT.

SIC RELOCATION

- PARSE TEXT RECORDS TO EXTRACT:
 - STARTING ADDRESS.
 - OBJECT CODE.
- ADJUST STARTING ADDRESS AND OBJECT CODE: ADD THE RELOCATION ADDRESS TO BOTH.
- OUTPUT THE UPDATED T AND E RECORDS:
 - T RECORDS CONTAIN RELOCATED ADDRESS AND OBJECT CODES.
 - E RECORD CONTAINS THE ADJUSTED EXECUTION START ADDRESS.



SICXE RELOCATION

- Parse modification records (M records) to determine:
 - Address requiring adjustment.
 - Whether to add or subtract the relocation address.
- Parse text records to:
 - Extract object codes.
 - Apply modifications based on M records.
- Output relocated T and E records.

ADDF = Add floating points,
SUBF = Subtract floating points,
MULF = Multiply floating points,
DIVF = Divide floating points

SCOFF FORMAT & STRUCTURES

1

H – header

2

T – text

3

M – modification

4

E – end record

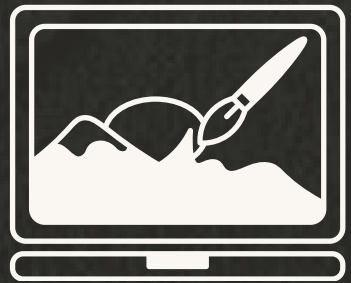
5

Parsed into
objFile

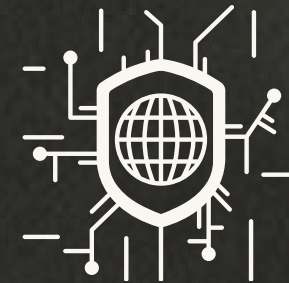
HIGH-LEVEL ARCHITECTURE

1. Main.c – argument parsing.
2. Loader.c – relocation pipeline.
3. ObjFileParser.c – parsing SCOFF.
4. RelocSic.c / relocSicXE.c – relocation logic.

OBJFILEPARSER - PURPOSE



- 01** Open object file
- 02** Identify record types



- 03** Parse fields
- 04** Validate format & addresses

PARSING DETAILS

- + TrimEolChars()
- + parseFixedHex()
- + Dynamic arrays (realloc)
- + Error flag pattern



LOADER PIPELINE



Main.c builds config

- Loader.c chooses SIC/SICXE backend
- Relocation applied to text bytes

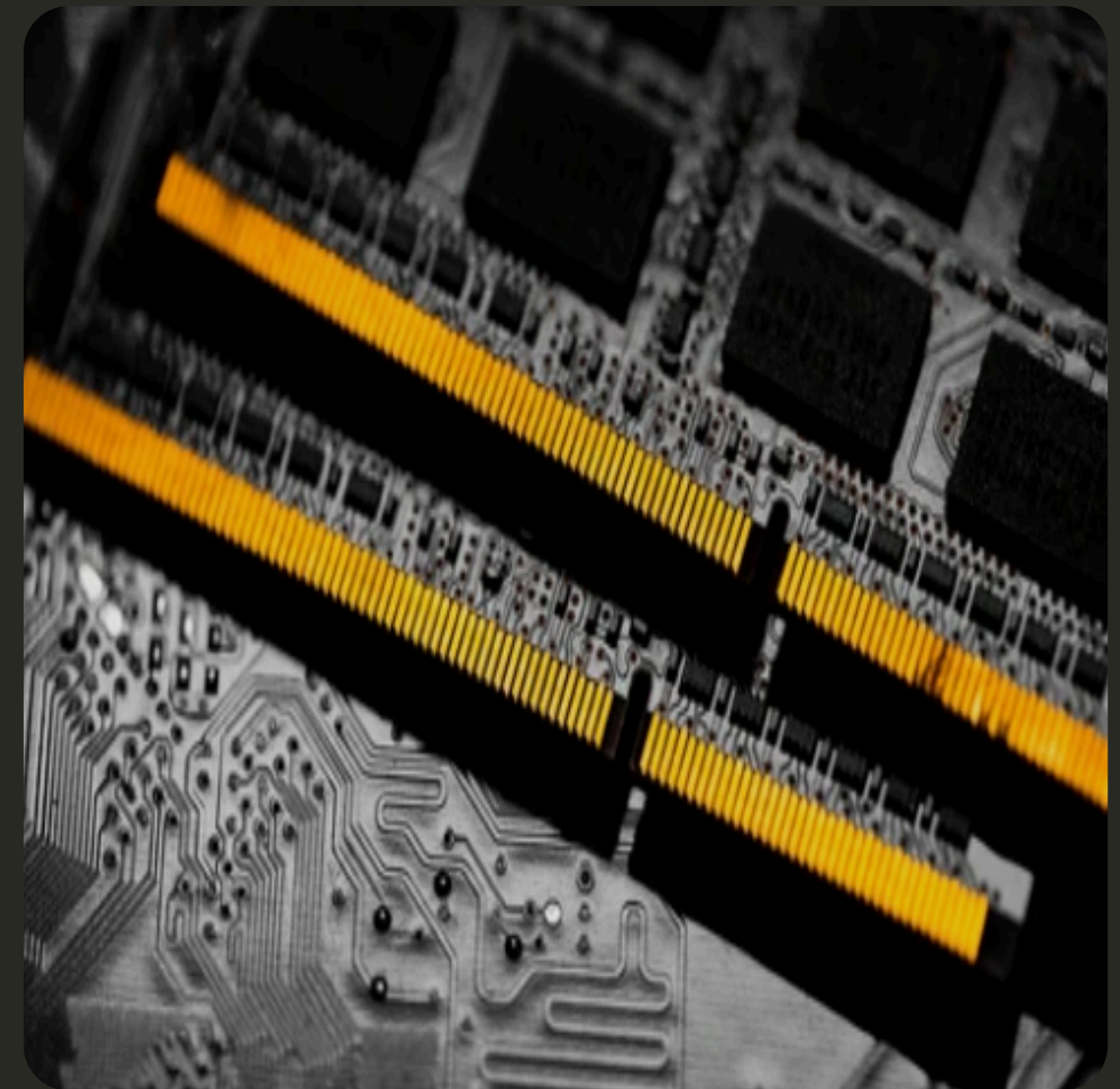
PROGRAM OUTPUT



Emit relocated T records



Update E record



CHALLENGES – PARSING



- ⊕ Strict SCOFF field widths
- ⊕ Handling malformed input
- ⊕ Avoiding goto for cleanup

CHALLENGES – RELOCATION

- ★ Nibble-level patching
- ★ Address range validation
- ★ Dynamic record management

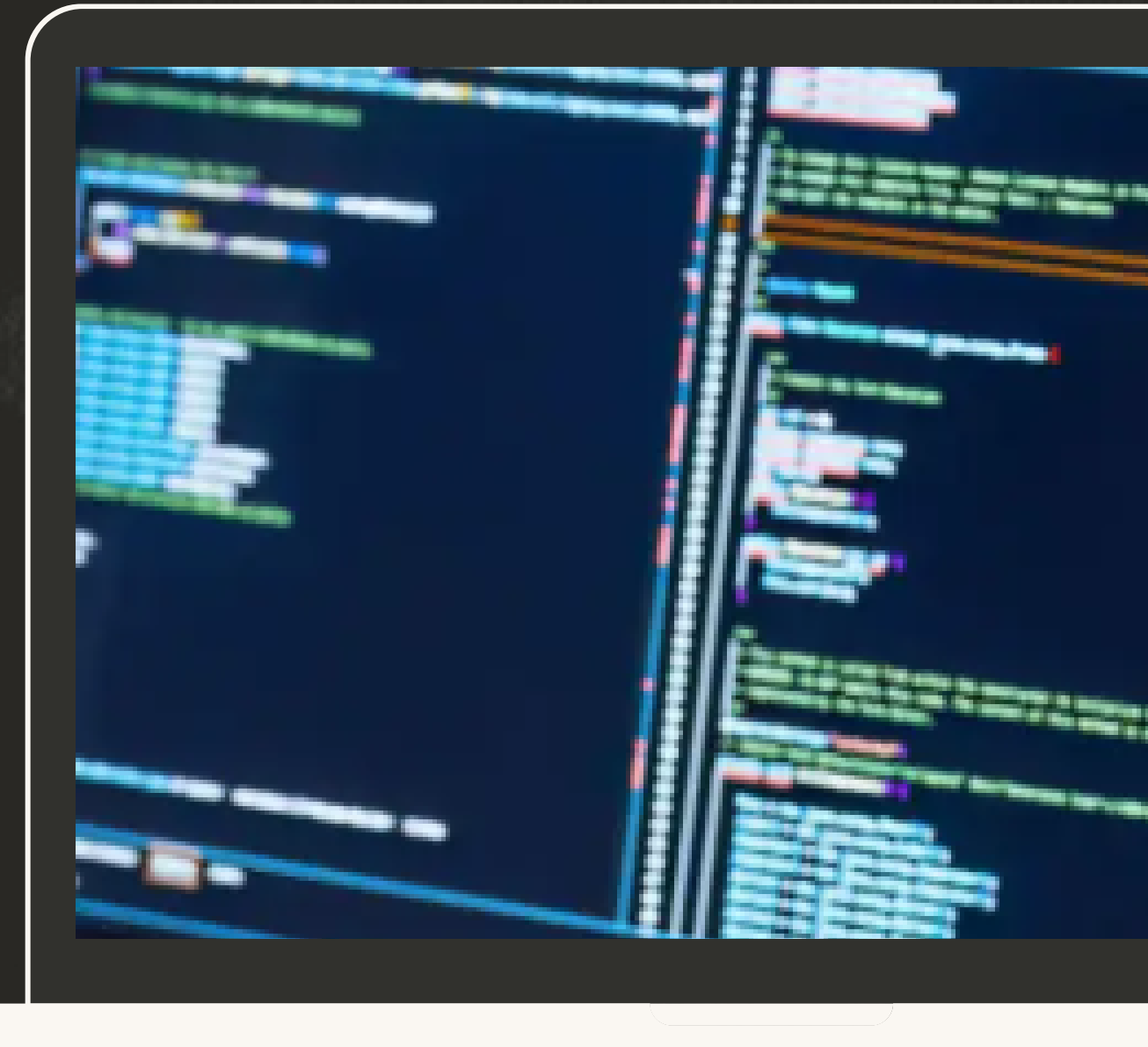
TESTING STRATEGY

The program accepts three arguments:

- ★ Object file name
- ★ Load address in hex
- ★ Machine type

Example:

- Command: `./ project5loader test1.sic.obj`
3500 SIC
- Explanation: Recolocate object file
“test1.sic.obj” to address “0x3500” for SIC.



CONCLUSIONS

- ★ Modular relocating loader
- ★ Robust parsing logic

THANK YOU